

MEMORANDUM

TO: Senior Operations Manager

FROM: Jackline Achieng' Mboya, Data Science Track

DATE: June 29, 2026

SUBJECT: Automation of Weekly Ward Efficiency Tracking & Rapid Exception Flagging

Executive Summary

This memo outlines a newly built digital monitoring tool designed to automate the tracking of weekly target goals across our hospital units. By shifting from manual spreadsheet reviews to an automated analysis pipeline, the department can instantly pinpoint underperforming facilities, protect staff from burnout, and ensure consistent patient care delivery.

Context: The Need for Automated Tracking

In busy healthcare environments, waiting until the end of the month to review operational logs exposes us to massive blind spots. Manual tracking requires coordinators to comb through raw spreadsheets, calculate percentages by hand, and cross-reference them against internal targets. This lag time means a unit could operate under severe strain for weeks before leadership notices a deficit. Automating this system guarantees real-time clarity with zero human calculation error.

How the Analytics Tool Operates (Plain English Breakdown)

The underlying system runs via a lightweight tracking program that operates on three core principles:

1. **The Formula Evaluator (Functions):** We created dedicated mathematical rules within the program that accept two pieces of information—the actual patient throughput and the target baseline—and immediately output an efficiency percentage.
2. **The Status Assigner (Conditional Logic):** The system automatically sorts units into clear action categories based on their score: "Normal" for meeting 90% or more of their targets, "Warning" for operational cracks (70%–89%), and "Critical" (below 70%) for immediate intervention.
3. **The Data Hub & Repeat Engine (Dictionaries & Loops):** We structured our clinic profiles into organized digital file cards. The program uses a rapid automated engine that cycles through every single profile in less than a second, applying the formula, assigning the status, and generating a clean, uniform report.

Operational Insight: Spotting Problems Faster

During our initial check on three test units, the tool instantly highlighted a critical variance:

- **The Maternal & Newborn Care Unit** and **Outpatient Emergency Clinic** are performing smoothly at 90% and 91.67% efficiency ("Normal").
- However, the **Pediatric Specialized Ward** registered an efficiency score of only 62.22%, triggering an immediate **"Critical"** alert.

In a manual system, this vulnerability would remain buried in numbers. With this tool, leadership immediately knows exactly where to allocate temporary staff or resources before patient care quality degrades.

Strategic Action & Deployment Recommendation

I recommend deploying this script directly into our regional operations reporting flow.

1. **Phase 1:** Sync this tool with our weekly intake ledger so that every Friday at 17:00 EAT, a standardized status report is auto-generated.
2. **Phase 2:** Connect the "Critical" trigger to an automated notification system that alerts department heads, cutting down response times from days to minutes.